# Retail Sales Analytics

Business Performance Analysis across Markets, Products, Pricing and Logistics

***Business Requirements Document***

## 1. Business Background

The business operated across 51,290 transaction records representing 25,035 orders from 795 customers between January 2011 and December 2014. Operations spanned seven markets (APAC, EU, US, LATAM, EMEA, Africa and Canada) and thirteen regions, generating $12.64M in sales and $1.47M in profit over the four-year period. The portfolio produced an overall profit margin of 11.62% and an average order value of $505.01.

Performance varied significantly across markets, products, discounts and shipping models. Similar levels of sales frequently translated into very different levels of profit, making company-wide averages insufficient for understanding the business.

The analysis was delivered through a six-page Power BI dashboard covering executive performance, markets and regions, products, customers and shipping, discounts and profitability, and business recommendations, supported by drillthrough and tooltip views.

## 2. Problem Statement

Between 2011 and 2014, revenue almost doubled from $2.26M to $4.30M. Profit margins, however, remained trapped inside a narrow 11-12% range across all four years (11.02%, 11.48%, 11.99%, 11.72%). The business was growing, but it was not becoming more profitable as it grew.

The company's 11.62% average margin made the business look more consistent than it actually was. Some parts of the business were converting revenue into profit extremely efficiently, while others were generating sales with little return or even outright losses.

This project was built to understand where those differences came from and whether they reflected isolated events or repeatable business patterns.

## 3. Data Scope

- Table: RetailSales -51,290 rows, 30 columns, one row per product line per order.
- Core fields: order_id, order_date, ship_mode, customer_name, segment, country, region, market, state, product_id, category, sub_category, product_name, sales, quantity, discount, profit, shipping_cost, order_priority, plus derived Discount Band and year/month fields.
- Period: January 1, 2011 – December 31, 2014 (4 full years).
- Geography: 7 markets, 13 regions, transactions spanning dozens of countries and states.
- Tools: MySQL for the query layer (25 queries), Excel for shaping, Power BI for the model and report.
- Data condition: zero missing values; one order_id can span multiple product lines, each with its own discount and profit.

## 4. Stakeholders

- Regional / market owners -sales rank and profit rank diverge by region; Southeast Asia and South both outsell several regions that out-earn them in profit.
- Category / merchandising owners -Furniture's revenue nearly matches Technology's without matching its margin, and one sub-category (Tables) loses money outright.
- Pricing / discount approval owners -profit turns negative at an identifiable discount level, not gradually.

- Logistics / fulfillment planners -shipping cost and delivery priority move independently of profit margin, sometimes in the opposite direction cost alone would predict.

## 5. Business Questions

**Q1 -Geography:** Central is the biggest business in the portfolio at $2.82M in sales, but its 11.03% margin sits close to the company average. Southeast Asia tells a very different story: $884K in sales turns into just $17.9K in profit and a 2.02% margin. Canada sits at the opposite extreme, generating only $66.9K in sales but converting it into the strongest margin in the dataset at 26.62%. Are the regions generating the most revenue also the regions creating the most value for the business?

**Q2 -Product:** Furniture generated $4.11M in sales against Technology's $4.74M, yet returned barely half the margin (6.98% vs 13.99%). Is this a Furniture problem, or is one small corner of the category making the entire business look weaker than it really is?

**Q3 -Discount:** Margins decline steadily from 25.32% without discounts to 9.86% at 10-20%, but the pattern changes completely beyond that point: -5.53% at 20-30% and -51.27% above 30%. Is there a point where discounts stop driving sales and start consuming profit instead?

**Q4 -Shipping / Fulfillment:** Average shipping cost rises from $19.97 in Standard Class to $42.94 in Same Day delivery. Should the business expect profitability to rise alongside those costs, or is something else driving margins entirely?

## 6. Analytical Requirements

**AR1 (→Q1):** Aggregate sales, profit and margin % by region (13) and by market (7); rank each by revenue and separately by profit, and compare the two rankings directly.

**AR2 (→Q2):** Aggregate sales, profit and margin % by category (3) and sub-category (17); isolate any sub-category with negative profit.

**AR3 (→Q3):** Bucket all 51,290 line items into 5 discount bands (0%, 0–10%, 10–20%, 20–30%, 30%+); compute sales, profit, margin % and line-item count per band.

**AR4 (→Q4):** Aggregate average shipping cost, average shipping days, sales and margin % by ship_mode (4) and separately by order_priority (4); compare cost/speed movement against margin movement in each.

## 7. Dashboard Requirements

**DR1 (→AR1):** Market & Regional Analysis page -a map (Global Sales & Margin Distribution), a "Highest Profit-Contributing Regions" bar chart, and a Market Revenue and Profitability combo chart, so revenue rank and profit rank sit side by side rather than being inferred from one sorted table.

**DR2 (→AR2):** Product Performance Analysis page -Category Revenue and Profitability combo chart, "Lowest Profit Margin Sub-Categories" bar chart, and a Sub-Category Portfolio Matrix scatter plotting margin against sales, so the one loss-making sub-category is visually isolated rather than buried in a 17-row table.

**DR3 (→AR3):** Discount & Profitability Analysis page -"Sales by Discount Band" and "Profitability by Discount Band" column charts placed together, plus KPI cards for Average Discount %, Revenue Under Discount, Profit Impact of Discounts, and % of Orders Discounted.

**DR4 (→AR4):** Customer & Shipping Analysis page -a "Shipping Cost vs Profit Margin by Ship Mode" scatter chart plotting the two variables directly against each other, plus a Delivery Performance gauge against a 3-day target.

# 8. KPI Definitions

| KPI | Formula | Dashboard Page |
| --- | --- | --- |
| Total Sales | Sum of sales | Executive Dashboard |
| Total Profit | Sum of profit | Executive Dashboard |
| Profit Margin % | Total Profit ÷ Total Sales | All pages (filtered by context) |
| Total Orders | Distinct count of order_id | Executive Dashboard |
| Average Order Value | Total Sales ÷ Total Orders | Executive Dashboard |
| Shipping Cost % | Total Shipping Cost ÷ Total Sales | Executive Dashboard |
| Revenue Growth % | (Last year sales − first year sales) ÷ first year sales | Executive Dashboard |
| Loss from 30%+ Discounts | Total Profit filtered to Discount Band = "30%+" | Discount & Profitability Analysis |
| Discounted Orders % | Distinct discounted orders ÷ distinct total orders | Discount & Profitability Analysis |
| Discounted Profit | Total Profit filtered to discount > 0 | Discount & Profitability Analysis |
| Unique Customers | Distinct count of customer_name | Customer & Shipping Analysis |
| Avg Revenue / Profit per Customer | Total Sales or Total Profit ÷ Unique Customers | Customer & Shipping Analysis |
| Avg Shipping Days | Average of Time for shipping, vs. 3-day target | Customer & Shipping Analysis |
| YoY Sales Growth % | (Current year sales − prior year sales) ÷ prior year sales | Business Insights & Next Steps |
| Profit Margin Change (pp) | Current year margin % − prior year margin % | Business Insights & Next Steps |

# 9. Key Discoveries

### Discovery 1 -The biggest markets are not necessarily the best businesses.

| Region | Sales | Profit | Margin % |
| --- | --- | --- | --- |
| Central | $2,822,399 | $311,404 | 11.03% |
| South | $1,600,960 | $140,356 | 8.77% |
| Southeast Asia | $884,438 | $17,852 | 2.02% |
| North Asia | $848,349 | $165,578 | 19.52% |
| Canada | $66,932 | $17,817 | 26.62% |

Southeast Asia outsells North Asia ($884,438 vs $848,349) but converts to roughly a ninth of the profit ($17,852 vs $165,578). Canada, the smallest region in the dataset by a wide margin, converts at the highest rate of any region (26.62%). The regions contributing the most revenue were often not the regions contributing the most profit.

### Discovery 2 -Furniture keeps pace with Technology on sales but falls far behind on profit.

| Category | Sales | Profit | Margin % |
| --- | --- | --- | --- |
| Technology | $4,744,691 | $663,779 | 13.99% |
| Furniture | $4,110,884 | $286,782 | 6.98% |
| Office Supplies | $3,787,330 | $518,474 | 13.69% |

Furniture almost matches Technology in sales but never comes close in profit. Most of that gap leads back to a single place: Tables, which sold $757K worth of products and still lost $64K over four years, at –8.47% margin.

### Discovery 3 -The data reveals a discount breakpoint rather than a discount curve.

| Discount Band | Sales | Profit | Margin % | Line items |
|---|---|---|---|---|
| 0% | $6,992,734 | $1,770,695 | 25.32% | 29,009 |
| 0–10% | $1,962,633 | $339,767 | 17.31% | 4,679 |
| 10–20% | $1,757,296 | $173,255 | 9.86% | 6,274 |
| 20–30% | $382,540 | –$21,156 | –5.53% | 967 |
| 30%+ | $1,547,702 | –$793,526 | –51.27% | 10,361 |

Between the 10–20% and 20–30% bands, margin falls roughly 15 points and turns negative. Past 30%, it collapses to –51.27%. That last band alone accounts for 10,361 of the dataset's 51,290 line items (20.2%) and –$793,526 in profit -larger in size than the business's entire recorded 2011 profit ($248,941).

### Discovery 4 -The orders costing the most to fulfill turned out to be the most profitable ones.

| Ship Mode | Avg Cost / Order | Avg Days | Margin % |
|---|---|---|---|
| Standard Class | $19.97 | 5.00 | 11.75% |
| Second Class | $30.47 | 3.23 | 11.46% |
| First Class | $41.05 | 2.18 | 11.37% |
| Same Day | $42.94 | 0.04 | 11.42% |
| **Order Priority** | **Avg Cost / Order** | **Avg Days** | **Margin %** |
| Critical | $59.72 | 1.81 | 12.76% |
| High | $32.87 | 3.09 | 11.04% |
| Medium | $18.44 | 4.52 | 11.87% |
| Low | $27.08 | 6.48 | 10.33% |

Shipping costs rose from $19.97 per order in Standard Class to $42.94 in Same Day delivery, yet profit margins barely moved from 11.75% to 11.42%. The bigger surprise appeared in order priorities: Critical orders cost $59.72 to fulfill on average -the highest in the business -and still produced the strongest margins at 12.76%. Low-priority orders told the opposite story, costing $27.08 per order compared to $18.44 for Medium priority, while returning the weakest margins at just 10.33%. The cost of serving an order and the profit earned from it were clearly not following the same pattern in this dataset.

## 10. Recommendations

**R1 (→Discovery 1):** Regional performance should be measured by how efficiently sales convert into profit, not by revenue volume alone. Southeast Asia behaves very differently from high-performing regions such as Central and deserves separate investigation rather than being treated as a smaller version of the same business.

**R2 (→Discovery 2):** Furniture looks like a weak category until opened up. Most of the damage comes from Tables. The rest of Furniture still makes money -just not at the level Technology or Office Supplies do.

**R3 (→Discovery 3):** The business seems comfortable living below 20% discounts. Above that line, it starts paying customers to buy products.

**R4 (→Discovery 4):** Paying more to move faster was rarely the problem in this business. Critical orders cost $59.72 to fulfill and arrived in 1.81 days, yet still generated the strongest margins at 12.76%. The weaker economics appeared at the other end of the queue, where Low-priority orders took 6.48 days to deliver, cost more than Medium priority orders ($27.08 vs $18.44), and returned the lowest margins in the portfolio at 10.33%.

# 11. Expected Business Impact

This section reflects what the historical data already contains, not a projection. The losses in this dataset are not spread evenly across the discount range. Most of them begin showing up beyond 20%. The 39,962 line items sold below that level stayed profitable across four years of data, while the 11,328 line items above it lost $814,682 - more than three times the company's entire profit in 2011. That comparison shows the size of a pattern already sitting inside four years of transaction data.

**This does not prove that limiting discounts would recover those losses. It does suggest that 20% behaves less like a pricing decision and more like a dividing line between two very different businesses.**